Chapter 12. Next Steps

It has been a long way since you were first introduced to counterfactuals. This book has taken you on a journey through the world of causal inference, starting with the basics and gradually building up to more advanced concepts and techniques. You should now have a solid understanding of how to reason about causation and how to use various methods to untangle causation from correlation in your data.

You have learned about the importance of A/B testing as the gold standard for causal inference, the power of graphical models for causal identification, and the use of linear regression and propensity weighting for bias removal. You have explored the intersection between machine learning and causal inference and how to use these tools for personalized decision making.

Furthermore, you have learned how to incorporate the time dimension into your causal inference analyses using panel datasets and methods like difference-in-differences and synthetic control. Finally, you have gained an understanding of alternative experiment designs for when randomization is not possible, such as geo and switchback experiments, instrumental variables, and discontinuities.

With the knowledge and tools presented in this book, you are equipped to tackle real-world problems and make informed decisions based on causation rather than correlation. I hope you enjoyed it and that it keeps being useful to you throughout your career.

This being an introductory book, I intentionally left out some of the causal inference topics that are active areas of research, but have not yet become widespread in the industry. This doesn't mean they aren't useful. Sometimes they are simply complicated, with no easy-to-use software that wraps them. If you enjoyed this book and you are craving something more, I suggest you explore one of the following topics.

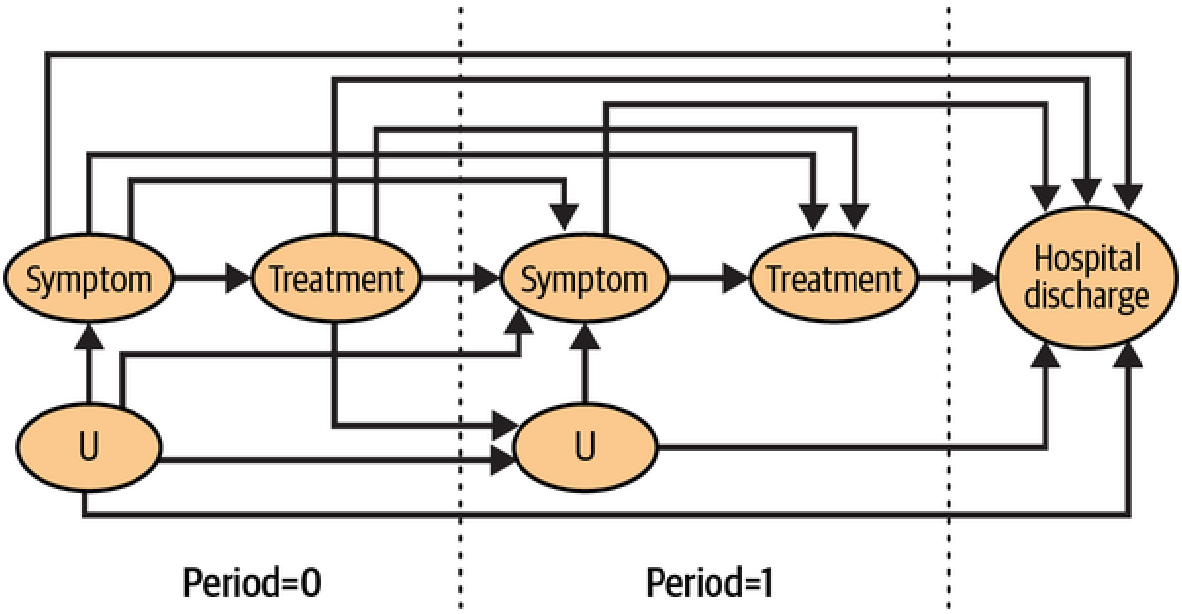
Causal Discovery

Throughout this book, you used causal graphs as a starting point for your causal inference analysis. But what if you don't know the causal graph and, instead, have to learn it from data? Causal discovery is a field of study that focuses on finding causal relationships between variables in a given system by using data generated from that system. Causal discovery is the process of going from data to causal knowledge. If you want to learn more about it, a good place to start is the paper "Causal Discovery Toolbox: Uncover

Causal Relationships in Python," by Diviyan Kalainathan and Olivier Goudet.

Sequential Decision Making

Although this book covered panel data structures, it did it mostly in the context of staggered adoption, which (among other things) means that there is no treatment-confounder feedback, which can typically arise when the treatment assignment is decided at each period in a sequence. To provide a concrete example, suppose you want to study the effect of a medical procedure (T) on hospital discharge rates (Y). However, the decision to perform the procedure depends on patient symptoms, and this decision is made on a daily basis. Therefore, the probability of a patient receiving the treatment on a particular day depends on whether they were treated on previous days and their symptoms over those days:



Although all the variables used to determine whether to treat or not are observable, traditional methods like regression may not be suitable for estimating the treatment effect due to the complex time dynamics and treatment-confounder feedback. Adjusting for confounders, such as patient symptoms, leads to noncausal paths, such as $T_0 \rightarrow Symptom_1 \leftarrow U_1 \rightarrow Y$.

Causal inference with sequential decision making has many applications in the industry. However, it is an incredibly intricate topic, which is why I left it out of this book. Still, if you are faced with a situation like the one I just described, I suggest you check out the book *Causal Inference: What If*, by Hernán and Robins. The last part of the book is dedicated to sequential decision making.

Causal Reinforcement Learning

Causal reinforcement learning (CRL) is an area of machine learning that combines the principles of causal inference and reinforcement learning. The goal of CRL is to automate the treatment allocation process with the objective of optimizing the outcome that the treatment influences. To achieve this goal, the automated decision-making system needs to balance exploiting promising treatments with exploring new treatments or applying the same treatment to different types of individuals. However, the use of observable variables in the decision-making process can lead to confounding, as there may be factors that affect both the treatment allocation and the observed outcome. Therefore, the system must adjust for these confounders to better understand the optimal treatments, which is a key challenge in CRL.

A simple example of where CRL could be applied is in the medical setting described earlier. However, instead of understanding the impact of a medical procedure, the objective would be to craft an agent that can recommend the procedure to physicians in a way that optimizes the patient outcomes. The agent would need to consider factors such as patient symptoms and medical history to make treatment recommendations that are tailored to each patient's individual needs while accounting for the causal relationships between the treatment and the observed outcomes.

Much of the literature in causal reinforcement learning gets entangled with that of contextual bandits. The two are, in fact, closely related. If you want a good place to start, I recommend the paper “Contextual Bandits in a Survey Experiment on Charitable Giving: Within-Experiment Outcomes Versus Policy Learning,” by Athey et al., and the American Economic Association Continuing Education Webcast on “Modern Sampling Methods,” by Keisuke Hirano and Jack Porter.

Causal Forecasting

Causal forecasting is a methodology that seeks to forecast future outcomes by taking into account the causal relationships between variables. Unlike traditional forecasting methods that rely solely on statistical associations between variables, causal forecasting aims to identify and model the underlying causal mechanisms that drive the relationships between variables. This approach can lead to more accurate and reliable forecasts, especially in complex systems where traditional statistical models may fail to capture the true causal relationships.

Causal forecasting typically involves a bit of causal discovery, since an important step in causal forecasting is figuring out if a correlation between X and Y is due to $X \rightarrow Y$, $Y \rightarrow X$, or $Y \leftarrow U \rightarrow X$. However, causal forecasting also requires dealing with the additional complexity of traditional time-series modeling, like nonstationarity and the data not being independent and identically distributed. A good place to learn more about this topic is the American Economic Association 2019 Continuing Education Webcast on Time-Series Econometrics, by James H. Stock and Mark W. Watson.

Domain Adaptation

Causal inference is the process of understanding what would happen from what did happen. This involves moving from a factual distribution, such as $Y|T = 1$, to a counterfactual one, like Y_1. The problem of inferring something about a distribution while having data from another one is known as *domain adaptation*, and it has many applications beyond causal inference. For instance, consider a financial services company that wants to detect fraudulent transactions. At first glance, this may appear to be a purely predictive task, where the company can train a machine learning model on its past transactions and use it to classify future transactions. However, the data the company has is fundamentally different from the data it needs to classify. Specifically, the company only has transactions that were authorized by its previous fraud-detection system. If that system was effective, then *P(fraud)* in the training data will be lower than *P(fraud)* for the future transactions the company model has to classify. In other words, the company has data on $Y|$ *filtered* but wants to build a model that is good at predicting *Y* without the filter. The company wants its model to act as the filter.

This is just one example, but there are many others. For instance, a company that is expanding into new countries may want to use its existing data from other countries to train predictive models that will perform well in the new country. Alternatively, a company's past data may behave differently from its current and future data, indicating that the distributions are shifting over time. In fact, since data is rarely stationary, most businesses will have to deal with distribution shifts in some way or another. This will require them to learn from one distribution to apply their insights to another. Although this problem is not strictly in the realm of causal inference, many of the techniques used in causal inference can be applied here. A good review on the literature on concept drift is given by the paper "Learning Under Concept Drift: A Review," by Lu et al.

Closing Thoughts

I hope I have sparked your interest in continuing your journey in causal inference. The nice thing about research is that it never ends. I myself intend to keep writing about causal inference for the foreseeable future and I would like to invite you to join me. You can find me on GitHub, Twitter, and LinkedIn, where I post regularly about causal inference. But most of all, my wish is that I have sparked in you an interest in this very fascinating topic. Although this book has come to an end, your learning journey on causal inference has just begun. I wish you all the best on the path ahead!